ModelIt!

Student Activity Pack

Why Your Phone Battery Dies So Fast

Energy Conversion, Efficiency, and the Heat Tax

Grade 09L1 | NGSS HS-PS3-1, HS-PS3-3 | Systems & Models

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What type of energy is stored in a phone battery? What does it convert into?

2 Why does your phone get hot when you play games or watch videos?

3 What does 'efficiency' mean in the context of energy conversion?

4 List three things that drain your phone battery the fastest.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Screen Brightness | Processing Load | Battery Capacity | Heat Generation | Battery Life

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Heavy Gaming

Screen Brightness: HIGH | Processing Load: HIGH

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Standby Mode

Screen Brightness: LOW | Processing Load: LOW

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Brightness Test

Processing Load: MEDIUM | Screen Brightness: LOW → HIGH

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Battery Age — How old the battery is measured in charge cycles. Lithium-ion batteries lose about 20% capacity after 500 full cycles.

Ambient Temperature — Environmental temperature affects battery chemistry. Extreme cold reduces available capacity; extreme heat accelerates degradation.

Background Apps — Programs running behind the scenes that consume power even when you're not using them — GPS, notifications, data syncing.

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why does your phone die in 3 hours of gaming but lasts all day on standby? Use evidence from your simulation to support your answers.

1 Why is it impossible to build a device with 100% energy efficiency?

2 How does the second law of thermodynamics relate to your phone getting hot?

3 What trade-offs do engineers make when designing phone power systems?

4 Why do older phones have worse battery life even with the same usage patterns?

5 How could understanding energy conversion help you design better technology?

8 STEM Challenge: The Battery Optimization Challenge ModelIt!

YOUR ENGINEERING MISSION

Design a power management system that maximizes battery life for a student who needs their phone to last a full school day (8 hours) with moderate use.

THE SCENARIO

A student's phone has a 4,000 mAh battery. They need to text, check social media during breaks, watch a 30-minute video at lunch, and use GPS to get home. Design power settings that keep the phone alive all day without carrying a charger.

Engineering Design Thinking

1. WHAT ARE THE HIGHEST-PRIORITY FUNCTIONS THE PHONE MUST PERFORM?
2. HOW WILL YOU SCHEDULE HIGH-POWER AND LOW-POWER ACTIVITIES ACROSS THE DAY?
3. WHAT TRADE-OFFS ARE YOU MAKING BETWEEN EXPERIENCE QUALITY AND BATTERY LIFE?
4. HOW WOULD YOU TEST WHETHER YOUR POWER PLAN ACTUALLY WORKS FOR 8 HOURS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Electrical Engineers design the power management systems in every device you own. They optimize energy conversion efficiency, design battery charging circuits, and create the algorithms that decide when to throttle performance to save power.